A Robust Framework for Improving Fine-Scale Evapotranspiration Estimation From UAV-Based Multispectral and Thermal Images

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***Abstract*—Uncrewed aerial vehicle (UAV)-based fine-scale evapotranspiration (ET) estimation is becoming increasingly critical in precision agricultural water management. However, existing UAV-based ET estimation studies often directly transfer satellite-based ET models and parameterization schemes to fine-scale UAV data, which hampers accurate fine-scale ET estimation. Here, we use machine learning (ML)-based alternative estimation schemes to estimate key parameters of aerodynamic roughness length (z_{0m}) and excess resistance (kB^{-1}) in the surface energy balance system (SEBS) ET model. In addition, we use a computational fluid dynamics (CFD) model to provide downscaled meteorological data for the SEBS model. Compared to physical parameterization schemes, ML-based estimates of z_{0m} and kB^{-1} show improved accuracy, reducing the mean root mean square error (RMSE) for z_{0m} from 0.07 to 0.04 m and for kB^{-1} from 4.58 to 2.41. Validation against eddy covariance (EC) systems with a source area of hundreds of meters shows that ML-based estimates of latent heat flux (LE) have an RMSE of 39.94 W/m^2, which is superior to the RMSE of 77.44 W/m^2 achieved by physical parameterization schemes. ML-based LE estimates also show comparable accuracy with an RMSE of 41.94 W/m^2 when using CFD-based meteorological data. A comparison with an optical-microwave scintillometer (OMS) system with a source area spanning kilometers confirmed the importance of CFD-based meteorological data and reduced the mean relative error (MRE) for LE from 26.53% (using site-observed meteorological data) to 22.28%. Our proposed robust framework improves the accuracy of UAV-based ET estimates, thus helping to bridge the scale gap between satellite remote sensing (RS) and site-based observations.**

***Index Terms*—Downscaled meteorological data, evapotranspiration (ET), machine learning (ML), uncrewed aerial vehicle (UAV).**

I. Introduction

As a key component of land hydrology, energy, and biogeochemical cycles, evapotranspiration (ET) plays an essential role in terrestrial water availability and consumption [1], [2]. Detailed knowledge of spatially distributed ET at a fine scale is especially beneficial for precision agricultural water management [3], [4]. This is particularly important because the reduced water supply in many water-scarce areas limits the ability to increase food production for the growing population [5]. The Food and Agriculture Organization (FAO) of the United Nations has argued that the pressures on land and water ecosystems are intense, and many of them are stressed at critical points [6]. ET is an important factor in water consumption; thus, its precise monitoring is of great significance for managing the risks to the quality of our land and water ecosystems under a changing climate and loss of biodiversity.

In the past several decades, numerous satellite optical and thermal remote sensing (RS) data have been used for ET estimation and monitoring [7], [8], [9]. Among them, widely documented ET products have pixel resolutions ranging from 30 m to 1 km, such as Landsat 8 imagery (30 m), MODerate resolution Imaging Spectrometer (MODIS) (250 m–1 km), and Sentinel 3 missions (500 m–1 km) [10], [11], [12]. To date, there is an urgent need for precision agricultural management at the field/subfield scale, but satellite data are insufficient to capture water status over a field. Recent advancements in near-surface RS technology have enabled uncrewed aerial vehicles (UAVs) to provide high spatial and temporal resolution observations of land surface status, bridging the scale gap between satellite RS and site-based observations [13], [14]. UAV data can be acquired at a fine spatial resolution of less than 1 m to monitor changes in plant physiology and ET

Received 20 January 2025; revised 25 June 2025; accepted 16 August 2025. Date of publication 21 August 2025; date of current version 23 September 2025. This work was supported by the National Key Research and Development Program of China under Grant 2023YFF1303503. *(Corresponding author: Shaomin Liu.)*

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Digital Object Identifier 10.1109/TGRS.2025.3601120

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under varying radiative, climatic, and hydrological conditions [14]. Recently, UAVs have been increasingly applied to derive spatially distributed ET and its components across diverse landscapes. These include grasslands [15], semiarid savannas [3], agricultural fields [16], urban environments [17], and vegetation mixtures in oasis–desert areas [18].

Given the above applications, thermal RS-based surface energy balance (SEB) models, such as SEB system (SEBS) [15], [18], two-source energy balance (TSEB) [4], and the dual temperature difference (DTD) model [16], have been widely adopted for UAV-based estimation of ET. UAV thermal imagery has relatively fine spatial resolution (<1 m), with each pixel typically representing a single land cover (LC) type [15]. One-source SEB models are well-suited for thermal imagery at submeter scales, as soil and vegetation canopy temperatures can be clearly distinguished. The one-source SEB model represented by the SEBS yields two key physical parameters: z_{0m} (aerodynamic roughness length) and kB^{-1} [19], [20], [21]. Although these parameters are physically based, their estimation remains a major source of uncertainty in SEBS applications. To address this issue, significant efforts have been made to refine their parameterization. Notably, the SEBS research group enhanced the representation of kB^{-1} over bare soil surfaces and proposed an improved scheme for kB^{-1} in forested environments [22], [23]. However, many UAV-based ET estimation studies tend to directly transfer these parameterization schemes to fine-scale UAV data [3], [4], [5]. Inappropriate parameterization schemes of z_{0m} and kB^{-1} have led to errors of 42% in estimating sensible heat flux (H) and 27% in estimating satellite-based latent heat flux (LE) [24], [25]. Therefore, there is a pressing need to develop new parameterization schemes tailored specifically for fine-scale multispectral and thermal UAV imagery.

SEB models require meteorological data, including air temperature (Ta), wind speed (Ws), and relative humidity (RH) to estimate ET [21], [26]. In satellite applications, these meteorological variables are typically retrieved from atmospheric reanalysis products. However, due to the coarse spatial resolution of atmospheric reanalysis products, most UAV-based ET estimates rely on site-based meteorological observations, under the assumption that meteorological conditions remain homogeneous throughout the flight [3], [4], [11], [16]. This dependence may limit the applicability of UAVs in fields lacking site-based meteorological observations. More importantly, site-based observations may not be representative enough when flight campaigns are conducted over large, heterogeneous areas with diverse underlying surfaces [13], [18]. Computational fluid dynamics (CFD) models have been widely adopted to estimate high-resolution atmospheric boundary layer flows and meteorological fields over heterogeneous landscapes [27], [28], [29]. This mechanism can be considered a dynamic downscaling process derived from atmospheric reanalysis data. Given their powerful and flexible simulation capabilities, CFD models offer a promising solution for providing the fine-scale meteorological data required by UAVs.

In recent years, machine learning (ML) methods have opened new opportunities for developing alternative estimation schemes. For example, Hu et al. [30] proposed a method for estimating z_{0m} and kB^{-1} using ML. Subsequently, Peng et al. [31] also applied ML methods to estimate z_{0m}, zero-plane displacement, and canopy height. Both studies focused on satellite-based pixels with coarse spatial resolution, but the proposed methods could be more effectively applied to fine-scale UAV-derived pixels. Against this background, we propose a robust framework for improving fine-scale ET estimation from UAV data. The objectives of this study are twofold: 1) to develop fine-scale, ML-based alternative estimation schemes for z_{0m} and kB^{-1}, coupled with CFD model-based downscaled meteorological data, and to validate them using site-based observations and 2) to evaluate the performance of UAV-based estimates of H and LE, using eddy covariance (EC) and optical-microwave scintillometer (OMS) observations. This study holds promise for improving the accuracy and applicability of UAV-based ET estimation.

II. Methodology

A. SEBS Model

This study employed the SEBS model, which has been widely used for UAV- and satellite-based RS estimation of water and heat fluxes [10], [15], [18], [21]. This model comprises three key components: 1) retrieval of physical parameters of the land surface; 2) estimation of surface roughness lengths for heat transfer (z_{0h}) and momentum transfer (z_{0m}); and 3) computation of the evaporative fraction (EF) and ET [20]. Model inputs include land surface parameters derived from UAV-based RS and meteorological data downscaled through a CFD model. Sensible heat flux (H) was calculated using the Monin–Obukhov similarity theory (MOST), while latent heat flux (LE) was derived from EF and available energy. The SEBS model can provide instantaneous fluxes at the time of each UAV flight. The required inputs for the SEBS model are illustrated in Fig. 1.

Similar to other one-source SEB models, the accuracy of SEBS relies heavily on the accurate estimation of the parameters z_{0m} and z_{0h} [22], [23]. The parameter z_{0m} defines the theoretical height at which wind speed drops to zero, whereas z_{0h} refers to the height at which the extrapolated Ta aligns with the actual surface temperature (Ts) [32]. The difference between z_{0m} and z_{0h} stems from the distinct processes governing heat and momentum transfer. While turbulent drag drives momentum transport, heat transfer is primarily influenced by thermal diffusivity and molecular viscosity [30]. To account for this difference, the excess resistance parameter kB^{-1} is introduced into the model. Their relationship can be expressed as follows:

$$z_{0h} = \frac{z_{0m}}{\exp\left(kB^{-1}\right)} \tag{1}$$

where k represents the von Kármán constant (0.4) and B denotes the Stanton number. kB^{-1} at each pixel can be derived by the equation provided by Su [20]

$$kB^{-1} = \frac{kC_d}{4C_t \frac{u*}{u(h)}\left(1 - e^{-\frac{n_{ec}}{2}}\right)} f_c^2 + 2f_c f_s \frac{k\frac{u*}{u(h)}\frac{z_{0m}}{h}}{C_t^*} + kB_s^{-1} f_s^2 \tag{2}$$

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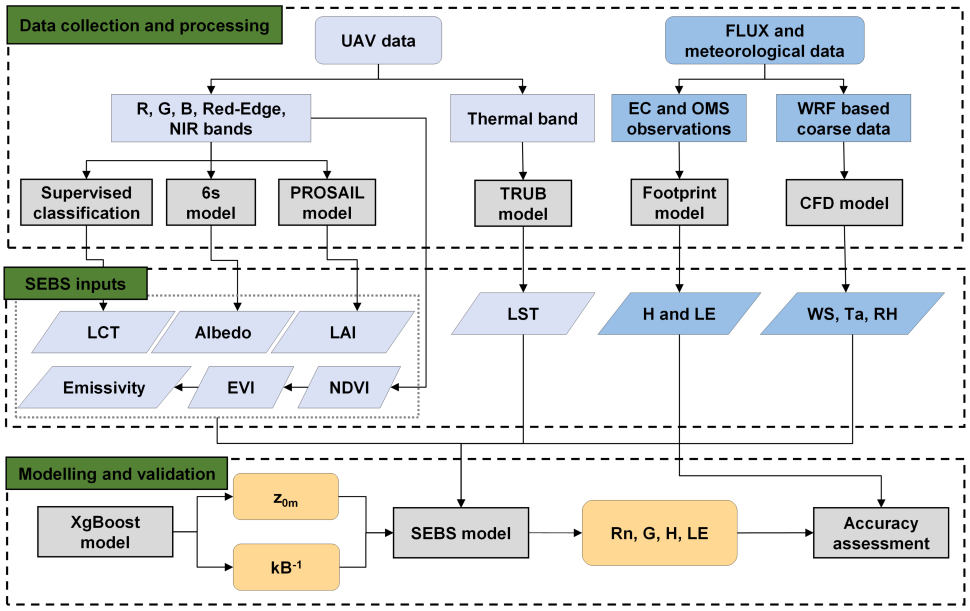

Fig. 1. Flowchart of the proposed framework for data collection, processing, inputs, modeling, and validation.

where C_d is the aerodynamic drag associated with the foliage, u_* is the friction velocity, and $u(h)$ is the wind speed at the canopy top. n_{ec} is the coefficient for wind speed extinction through the canopy, h is the canopy height, and C_t^* is the soil thermal transfer coefficient. f_c is the fractional canopy coverage, and f_s is the fraction of bare soil in one pixel. kB_s^{-1} corresponds to the kB^{-1} for bare soil. The calculations of these parameters can be found in [20].

The surface roughness lengths for momentum transfer z_{0m} are determined as follows:

$$z_{0m} = h\left(1 - \frac{d}{h}\right)e^{-ku(h)/u_*}. \tag{3}$$

The zero-plane displacement height (d) is calculated based on the canopy height (h) and the wind speed extinction coefficient. The calculation of z_{0m} is also provided in [33].

More general information about the SEBS model and detailed physical parameterization schemes can be found in [20]. Due to the relatively low vegetation height across the study area, we adopted the SEBS version developed by Chen et al. [22], which features an optimized parameterization of the excess resistance to heat transfer for bare soil surfaces. Given their importance in the SEBS model, we aim to develop ML-based estimation schemes for z_{0m} and kB^{-1} when the model is applied to UAV-based fine-scale imagery. The detailed methods are presented in Section II-B.

B. ML-Based Estimation Schemes

ML, as a new data-driven technique, has received increasing attention in the fields of RS and Earth science over the past decade [30], [34]. ML has demonstrated its ability to accurately predict the state of complex systems and provide better descriptions of physical processes based solely on data [30], [35], [36]. Therefore, we propose alternative estimation schemes using the ML methods to estimate z_{0m} and kB^{-1}. The extreme gradient boosting (XGBoost) model is adopted due to its high accuracy and computational efficiency and has also been widely adopted for the simulation of ET, soil moisture, and ecosystem respiration [37], [38], [39]. XGBoost is a scalable gradient boosting model based on an ensemble of classification and regression trees (CARTs). It was first proposed by Chen and Guestrin [40], and like most gradient boosting models, it was also developed based on a residual optimization algorithm. XGBoost is designed with parallelization for computing feature gain, depth-first tree pruning, and algorithmic enhancements to ensure high accuracy and increase training speed.

The construction process of the alternative estimation schemes using the XGBoost model primarily consists of two parts: the selection of input variables and the determination of the model hyperparameters. Some studies have attempted to use ML models to estimate z_{0m} or kB^{-1} in satellite RS-based applications [30], [31]. Due to the limited coverage area and pure pixel characteristics of UAVs, the application of ML effectively avoids predictions outside the training samples and reduces errors caused by mixed pixels. For feature selection, we chose only the input variables used in previous studies for model training [30], [31]. Specifically, the LC type, the day of year (DOY), the normalized difference vegetation index (NDVI), and the enhanced vegetation index (EVI) were selected as input variables following the method of Peng et al. [31]. Notably, elevation-related information was not included although it was suggested by Peng et al. [31] because the study area has relatively flat topographic conditions. In contrast to z_{0m}, kB^{-1} exhibits distinct diurnal variation characteristics [41]. The LC, DOY, land Ts, Ta, RH, Ws, NDVI, and EVI were selected as input variables for the ML-based kB^{-1} estimation [30]. In addition to the diurnal variation characteristics that included Ts, Ta, RH, and Ws at half-hour intervals, clock information (labeled from 1 to 48 in 30-min intervals throughout the day) was also used in this study. The solutions for z_{0m} and

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kB^{-1} using the XGBoost model can be expressed as follows:

$$z_{0m} = \text{XGBoost}\,(\text{LC, DOY, NDVI, EVI}) \tag{4}$$

$$kB^{-1} = \text{XGBoost}\begin{pmatrix}\text{LC, DOY, Ts, Ta, RH, Ws,}\\ \text{NDVI, EVI, clock}\end{pmatrix}. \tag{5}$$

The observed z_{0m} can be calculated by inverting the wind profile equation [31], using the mean wind speed and turbulent flux values measured by the sonic anemometers of the EC system. The equation is presented as follows:

$$z_{0m} = \frac{z-d}{\exp\left(\frac{ku}{u_*}\right) + \Psi_m\left(\frac{z-d}{L}\right)} \tag{6}$$

where u_* is the friction velocity, z is the measurement height of the wind speed above ground, and L is the Obukhov length. Ψ_m is the stability correction function for momentum transfer. In this equation, three parameters of d, u_*, and L are required to characterize z_{0m}. Generally, EC observations include u_* and L. To eliminate d in the inversion of the wind profile equation, the equation $d = 6.67\ z_{0m}$ is used [31]. The observed z_{0m} is assumed to remain constant for shelterbelts and built-up areas. For grasslands and croplands, z_{0m} is assumed to remain constant within a seven-day running window and within a 15-day running window for other LC types. The robust estimates of z_{0m} can be obtained by minimizing the residuals during a certain period with the cost function in the wind profile equation. A more detailed calculation process can be found in [31].

The half-hourly kB^{-1} observations are inferred based on the temperature profile equation [30], using the temperature values and turbulent flux measurements by the automatic weather station (AWS) and the EC system. The expression is described as follows:

$$kB^{-1} = \frac{ku_*\rho c_p\,(T_s - T_a)}{\mathrm{H}} - \ln\left(\frac{z-d}{z_{0m}}\right) + \Psi_h\left(\frac{z-d}{L}\right). \tag{7}$$

where ρ is the air density, c_p is the specific heat of air, and Ψ_h is the stability correction function for H.

The hyperparameters were determined through grid search and iterative tuning, as each ML model requires dataset-specific configurations. Given the 1-D model input used in this study, model complexity was moderate, and fewer hyperparameters required optimization. Based on parameter settings from previous ML-based studies [38], [40], we optimized six hyperparameters: the number of estimators (n_estimators), learning rate (learning_rate), minimum split loss (gamma), subsample ratio (subsample), column sample by tree (colsample_bytree), and maximum depth (max_depth). Their respective search ranges were {1000, 8000}, {0.01, 0.20}, {0, 5}, {0.5, 1.0}, {0.5, 1.0}, and {3, 10}. Based on tenfold cross-validation, the optimized hyperparameters of n_estimators, learning_rate, gamma, subsample, colsample_bytree, and max_depth were set to 2000, 0.01, 0, 0.8, 0.9, and 10 for z_{0m}, respectively, and 7000, 0.01, 0, 0.8, 0.9, and 6 for kB^{-1}, respectively. Notably, the ML model for z_{0m} was trained at the daily scale due to its temporal stability, whereas the ML model for kB^{-1} was trained at the half-hourly scale because of its diurnal variation. Once these two models are trained using site-based observations, the spatially distributed z_{0m} and kB^{-1} can be estimated with UAV-based land surface parameters and downscaled meteorological data.

C. CFD Model-Based Downscaling Method

The CFD model used in this study was employed to generate high-resolution, spatially distributed meteorological data [28], [29]. The use of the CFD model in this study serves two primary purposes. First, it reduces the reliance on site-based meteorological observations, making it possible to conduct UAV campaigns without additional meteorological measurements. Second, it explicitly accounts for the complexity of meteorological conditions, particularly when site-based observations may not adequately represent heterogeneous flying areas. The widely used CFD model calculates the wind, Ta, and humidity fields iteratively following the principles of mass, momentum, and energy conservation. In addition, customized modules are incorporated to account for radiation, the soil energy balance, and the vegetation energy balance, providing a detailed description of energy and water vapor exchange within vegetation. Specific source terms are incorporated into the governing equations to reflect the impacts of vegetation and buoyancy forces [28]. Specifically, the numerical simulation of atmospheric boundary layer flows is conducted using the standard k–ε turbulence closure model coupled with incompressible Reynolds-averaged Navier–Stokes (RANS) equations. The influence of vegetation on airflow patterns is represented through a porous medium approach, which is incorporated into the governing equations as a source term to capture the interaction between land surface (soil and vegetation) and atmosphere. Slip boundary conditions are applied at the top and lateral boundaries of the domain, while the bottom surface is treated as a no-slip wall boundary and characterized using the rough wall law formulation. A fixed velocity is prescribed at the inlet, and a constant static pressure is set at the outlet to ensure stable and flow development. More detailed information about the CFD model can be found in [29].

Given its powerful capabilities, the CFD model has been widely used for simulating urban heat island circulation and oasis–desert interactions [28], [42]. In general, the CFD model offers unique insights for dynamically downscaling coarse atmospheric reanalysis data to finer scales. Unlike statistical downscaling, dynamical downscaling is a physical method that generates fine-scale meteorological data [43]. For this study, atmospheric reanalysis data generated by the weather research and forecasting (WRF) model with a spatial resolution of 3 km were used as inputs for the CFD model [38]. The outputs include spatially distributed Ws, RH, and Ta at a spatial resolution of 30 m and a temporal resolution of 1 h.

III. Study Area and Data

A. Study Area and Experiments

The Heihe River Basin (HRB) is the second-largest endorheic basin in the arid and semiarid regions of Northwest China, covering approximately 143 000 km^2 [44]. The

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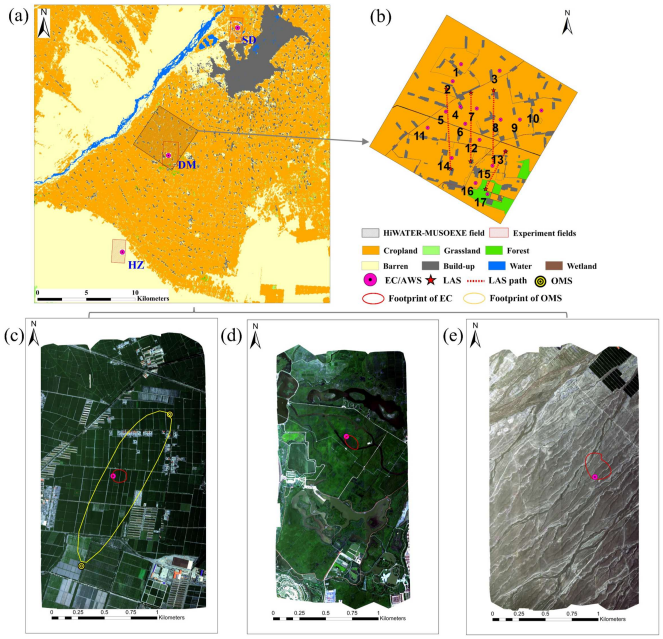

Fig. 2. (a) Study area in the midstream of HRB, (b) intensive flux matrix of HiWATER-MUSOEXE field and associated observatory sites, and the locations of UAV experimental fields in (c) DM, (d) SD, and (e) HZ, respectively.

landscapes within the HRB are diverse, with ice and snow, alpine meadows, and Qinghai spruce predominantly found in upstream areas, artificial oases (i.e., croplands and shelterbelts) surrounded by deserts in the midstream area, and deserts interspersed with small natural oases (i.e., riparian forests) located primarily downstream. The study area is situated in the midstream region of the HRB, which is predominantly composed of irrigated agricultural landscapes and shelterbelts surrounded by desert [see Fig. 2(a)]. This oasis represents the largest along the Hexi Corridor and serves as a typical irrigated crop ecosystem, encompassing maize fields, vegetable farms, orchards, shelterbelts, residential areas, wetlands, water bodies, and deserts [18]. The Heihe Integrated Observatory Network has provided the spatiotemporal dynamics of land surface fluxes, hydrometeorological parameters, vegetation, and other factors over the whole HRB [45].

To date, there are a total of 15 observatory sites across the entire HRB, representing the main LC types. In the midstream, three sites are located in and around the oasis–desert area [see Fig. 2(a)]. The three stations are the Daman superstation (DM), the Zhangye wetland station (SD), and the Huazhaizi desert steppe station (HZ). All of them provide meteorological and flux observations through AWSs and EC systems. Notably, an intensive flux matrix experiment named Multi-Scale Observation Experiment on ET over Heterogeneous Land Surfaces (HiWATER-MUSOEXE) was conducted from May to September 2012. The flux observation matrix consisted of two nested matrices: one large experimental area (30 × 30 km) in an oasis–desert area and one kernel experimental area (5.5 × 5.5 km) within the Zhangye oasis [see Fig. 2(b)]. In the 5.5 × 5.5 km oasis region, 17 measurement sites were established, each equipped with AWS, EC, and four

TABLE I
INFORMATION ON UAV FLIGHT CAMPAIGNS

Region	Size (km^2)	Date	Ws (m/s)	Ta (°C)	RH (%)	Land cover
DM	2.7×1.6	2019/5/27	2.25	19.07	26.33	Maize orchard shelterbelt road build-up
		2019/6/29	1.11	22.38	57.46	
		2019/7/22	1.81	24.76	52.21	
		2019/8/23	1.62	19.58	91.00	
		2019/9/24	1.49	20.21	43.99	
		2020/6/18	2.45	28.87	28.22	
		2020/7/13*	0.60	24.22	71.66	
		2020/8/20	1.66	28.12	39.96	
		2020/9/19	1.33	24.17	33.39	
SD	2.4×1.2	2019/5/28	2.42	20.20	24.70	Reed water road build-up
		2019/6/30	1.76	25.58	31.98	
		2019/7/25	4.08	24.62	47.16	
		2019/8/24	2.73	20.76	33.45	
		2019/9/23	2.12	21.87	30.83	
		2020/6/17*	1.08	28.11	33.31	
		2020/7/14	1.08	28.11	33.31	
		2020/8/21	2.50	23.29	58.64	
		2020/9/20	4.39	18.07	45.23	
		2020/10/21	2.42	20.20	24.70	
HZ	2.4×1.4	2019/5/26	2.63	12.03	12.90	Desert *Kalidium foliatum* maize
		2019/6/28	1.88	20.45	51.07	
		2019/7/24	1.75	22.43	48.09	
		2019/8/22	1.86	18.81	74.14	
		2019/9/22	1.45	19.49	37.82	
		2020/6/14	2.79	24.42	28.82	
		2020/7/14	3.00	29.40	26.61	
		2020/9/19	3.41	25.41	27.26	
		2020/10/20	2.62	9.55	31.78	

* indicates the advection occurred during the flying campaign.

sets of large aperture scintillometers (LASs), among other instruments [46].

The UAV experiment, known as the MUltiScale Observation Experiment on land Surface ET-UAV (MUSOES-UAV), was conducted across DM, SD, and HZ, which are part of the Heihe integrated observatory network [see Fig. 2(c)–(e)] [47], [48]. Flight campaigns were conducted monthly during the growing seasons of 2019 and 2020, specifically from May to September 2019 and June to October 2020. The experimental fields comprised three rectangular areas, each measuring 2 × 1 km, strategically established to align with the MODIS pixels and the source regions of EC systems. The DM experimental field primarily featured maize, orchards, and shelterbelts interspersed with roads and residential zones. The SD experimental field was characterized by wetlands, including reeds, water bodies, roads, and residential areas. The HZ experimental field consisted mainly of desert terrain dominated by *Kalidium foliatum*. More detailed information about the UAV experiment can be found in Table I.

B. UAV System and Data

Two DJI M600 Pro UAVs that labeled UAV1 and UAV2 were deployed to collect multisensor data across the three experimental fields. UAV1 was equipped with a Micasense RedEdge-M multispectral camera and a thermal imager, enabling simultaneous capture of visible–near-infrared (VNIR) and thermal infrared (TIR) imagery. The multispectral camera

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recorded five bands, blue, green, red, red-edge, and near-infrared, while the thermal imager operated within a spectral range of 7.5–13.5 μm. The spatial resolutions for VNIR and TIR imagery were approximately 0.2 and 0.4 m, respectively. UAV2, equipped with a Tovos DroneScan LiDAR system, generated 3-D point cloud data of the terrain. Data acquisition primarily occurred between 10:00 and 14:00 Beijing Time (BJT), aligning with the overpass schedules of the MODIS Terra and Aqua satellites.

For UAV data preprocessing, several steps were followed based on the methods described by Wei et al. [18]. Raw VNIR and TIR images collected during UAV flights were imported into Pix4Dmapper software. The software was then used to generate a digital surface model (DSM) and orthophotos. The DSM was integrated with the VNIR and TIR orthophotos to correct for relief displacement. The VNIR orthophotos were converted into five-band reflectance data, and the TIR orthophotos were processed to obtain the surface brightness temperature.

To obtain specific land surface parameters in our proposed framework, the second simulation of the satellite signal in the solar spectrum (6S) model (version 4.1) was applied. This model is used to convert narrowband albedo into broadband albedo. The leaf area index (LAI) was derived using the PROSAIL model based on VNIR images, along with the gap fraction theory from LiDAR data. The NDVI threshold method was used to calculate land surface emissivity [49]. In addition, Ts was estimated using the temperature retrieval for UAV broadband (TRUB) method, which was applied to the surface brightness temperature [47], [48]. Validation with site-based observations confirmed the robustness of these retrieved parameters. For more details on the retrieval methods and validation, refer to Wei et al. [18]. LC classification was performed using eCognition software (version 9.0). This classification process is essential for effectively training the ML models.

C. Meteorological and Flux Data

Among these three experimental fields, the meteorological elements, radiation, water, and heat flux observations were provided during each flying campaign. Specifically, the AWS measured wind speed/direction, Ta and humidity, atmospheric pressure, radiation, soil heat flux, precipitation, soil temperature, and moisture profile observations. The H and LE observations from the EC system were provided simultaneously during the study period. The original EC data were processed with the raw 10-Hz data, and the 30-min average EC data values were calculated using EddyPro software. The source areas of EC were estimated by the Eulerian analytic flux-footprint model at a scale of approximately hundreds of meters [50], [51].

The DM station also collected H and LE data using an OMS in 2020. The source areas for the OMS measurements were defined using the footprint model for point fluxes, combined with the path-weighting function of the OMS, which covers an optical path length of 1854 m. This corresponds to an approximate scale of 2 × 1 km. Detailed data processing methods for the OMS measurements are outlined by Zheng et al. [52]. In addition to the three UAV campaign sites, site-based observations from HiWATER-MUSOEXE were utilized in this study. These additional sites were included to ensure an adequate number of samples for training z_{0m} and kB^{-1}. Detailed information about the observatory sites used in the model training is presented in Table II.

TABLE II
SUMMARY OF OBSERVATORY SITES TO PROVIDE METEOROLOGICAL AND FLUX OBSERVATIONS

Site name	Year	Lon. (E)	Lat. (N)	Duration	Land cover
DM	2018-2021	100.3722	38.8555		Cropland
SD		100.4464	38.9751	5/20-10/21	Wetland
HZ		100.3186	38.7652		Desert
site 1	2012	100.3582	38.8932	6/10-9/17	Cropland
site 2		100.3541	38.8870	5/3-9/21	Cropland
site 3		100.3763	38.8906	6/3-9/18	Cropland
site 4		100.3575	38.8775	5/10-9/17	Build-up
site 5		100.3506	38.8758	6/4-9/18	Cropland
site 6		100.3597	38.8712	5/9-9/21	Cropland
site 7		100.3652	38.8768	5/28-9/18	Cropland
site 8		100.3765	38.8725	5/14-9/21	Cropland
site 9		100.3855	38.8724	6/4-9/17	Cropland
site 10		100.3957	38.8757	6/1-9/17	Cropland
site 11		100.3420	38.8699	6/2-9/18	Cropland
site 12		100.3663	38.8652	5/10-9/21	Cropland
site 13		100.3785	38.8608	5/6-9/20	Cropland
site 14		100.3531	38.8587	5/6-9/21	Cropland
site 15 (DM)		100.3722	38.8555	5/10-9/26	Cropland
site 16		100.3641	38.8493	6/1-9/17	Cropland
site 17		100.3697	38.8451	5/12-9/17	Orchard
BJT		100.3042	38.9150	5/13-9/21	Desert
SSW		100.4933	38.7892	6/1-9/21	Desert
HZ		100.3186	38.7652	6/12-9/21	Desert
SD		100.4464	38.9751	6/25-9/21	Wetland

To ensure high-quality data for training the z_{0m} and kB^{-1} parameters, a rigorous data screening process was applied in this study. The UAV campaign spanned from May to October, and only daytime observations (9:00–16:00) were selected for model training. Data collected during instrument malfunctions, maintenance periods, and within 30 min before or after precipitation events were excluded. Each 30-min flux dataset was assigned a quality flag according to the quality assessment methods including stationarity and integral turbulent characteristics tests, and low-confidence data were discarded. Following this initial screening, additional criteria were applied to the meteorological and flux observations. Outliers beyond three standard deviations ($3-\sigma$) were excluded, and data with friction velocities below 0.01 m/s and wind speeds below 1 m/s were removed, in line with the approach of Hu et al. [30]. Rows with stability correction functions (ς) outside the range from −2 to 1 were also excluded [53]. For kB^{-1}, data with H values below 10 W/m^2 and temperature differences between Ts and Ta less than 0.1 °C were removed to mitigate the effects of advection [30]. It should be noted that data from all sites listed in Table II were used in the training process to better generalize the trained model across each 2 × 1 km study area. Specifically, 75% of the data from each site were used for training, and the remaining 25% were used for validation. In addition, tenfold cross-validation was applied to further evaluate the performance of the ML model and ensure its robustness and reliability.

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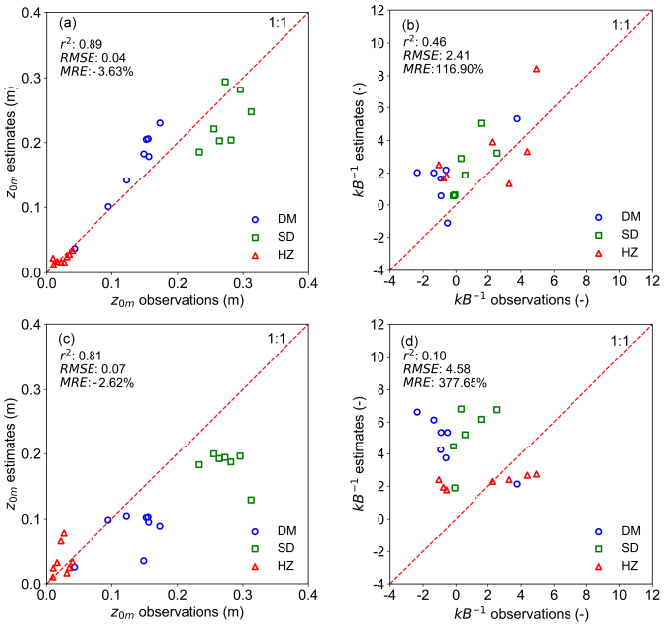

Fig. 3. Validation of ML-based estimates of (a) z_{0m} and (b) kB^{-1}, as well as their corresponding estimates (c) and (d) derived from the physical parameterization schemes of the SEBS model.

D. Auxiliary Data

In addition to meteorological and flux data, vegetation-related NDVI and EVI data were utilized for model training. The ML model was trained at the source area of flux observations, which covers hundreds of meters. Therefore, the NDVI and EVI pixels within the source area were extracted from the Landsat satellites using the Google Earth Engine (GEE) planetary-scale platform. The Landsat-based NDVI and EVI products have the spatial and temporal resolutions of 30 m and eight days, respectively. To match the required daily scale, both datasets were smoothed and interpolated to daily time series following Chen et al. [54]. Furthermore, coarse-scale atmospheric reanalysis data (Ws, Ta, and RH) generated by the WRF model were incorporated into this study [38]. These meteorological data have the spatial and temporal resolutions of 3 km and 1 h, respectively. The spatial resolution was improved to 30 m using the CFD model.

IV. Results

A. Validation of Estimated Key Parameters and Downscaled Meteorological Data

Fig. 3 presents the validation results of the ML-based estimates of z_{0m} and kB^{-1} against the corresponding site-calculated observations by (6) and (7). It is important to note that pixels within the source area of EC were extracted, and footprint-weighted average values were used during validation. In general, the ML method agreed well with site-based observations. In terms of r^2, the z_{0m} and kB^{-1} estimates were significantly correlated with their respective observations [see Fig. 3(a) and (b)]. In addition, the physical parameterization schemes of z_{0m} and kB^{-1} from Chen et al. [22] were included for comparison purposes [see Fig. 3(c) and (d)].

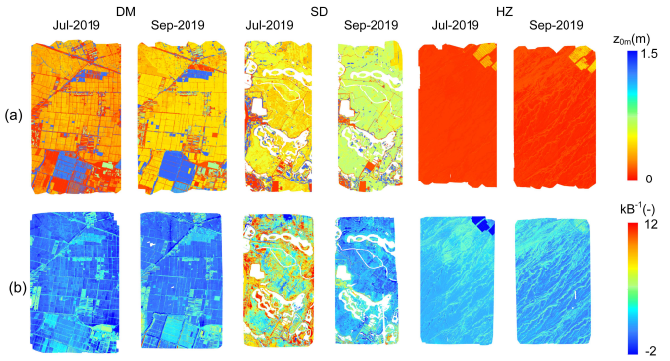

Fig. 4. Spatial distributions of ML-based estimation of (a) z_{0m} and (b) kB^{-1} for DM, SD, and HZ experimental fields.

The ML method improved the performance of z_{0m} estimates, reducing the root mean square error (RMSE) from 0.07 to 0.04 m and the mean relative error (MRE) from −3.63% to −2.62%, compared with physical parameterization schemes. This improvement was more obvious in DM and SD experimental fields, where roughness elements were generally greater than those in the HZ experimental field. Compared with the physical parameterization schemes, ML-based kB^{-1} estimates improved overall accuracy, with the RMSE decreasing from 4.58 to 2.41 and the MRE improving from −377.65% to −116.90%.

Fig. 4 presents the spatial distributions of z_{0m} and kB^{-1} estimated by the ML method, with values ranging from 0 to 1.5 m and −2 to 12, respectively. In general, noticeable differences within the same experimental field can be clearly identified. High values of z_{0m} were observed in croplands, shelterbelts, residential areas, and reeds [see Fig. 4(a)]. In the HZ experimental field dominated by desert, the z_{0m} estimates were nearly zero, with the exception of those for croplands located in the northeast corner of the field. A comparison of z_{0m} between July and September also revealed differences related to growth stages, with higher z_{0m} values observed in September than in July. Similarly, kB^{-1} exhibited spatial distributions opposite to those of z_{0m}, showing distinct patterns influenced by underlying surfaces. High values of kB^{-1} were predominantly found in nonvegetated areas such as residential areas, roads, and deserts.

Fig. 5 presents scatterplots between downscaled meteorological data and site-based observations. The downscaled Ws, Ta, and RH from the CFD model showed good agreement with site-based observations. The coefficient of determination (r^2) indicated that the downscaled meteorological data had a correlation of at least 0.68 with site-based observations, particularly high r^2 values of 0.91 for Ta and 0.86 for RH. The RMSE values for the Ws, Ta, and RH estimates were 0.63 m/s, 1.47 °C, and 5.87%, respectively. The estimates of wind speed were slightly underestimated, with an MRE of −12.92% compared to site-based observations. This underestimation was primarily observed in the HZ and SD experimental fields under high wind speeds (see Table I). In addition, a slight overestimation of Ta was also observed during several cold days, with an MRE of 4.37% [see Fig. 5(b)]. In contrast, RH

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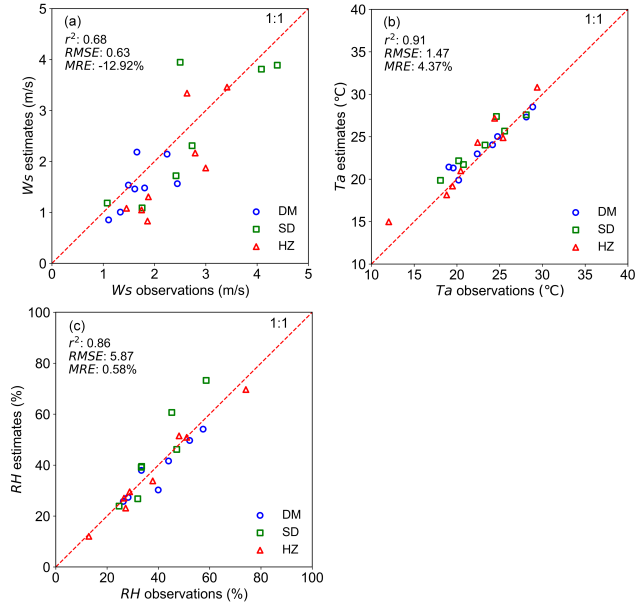

Fig. 5. Validation of CFD model-derived (a) wind speed, (b) Ta, and (c) RH with site-based observations.

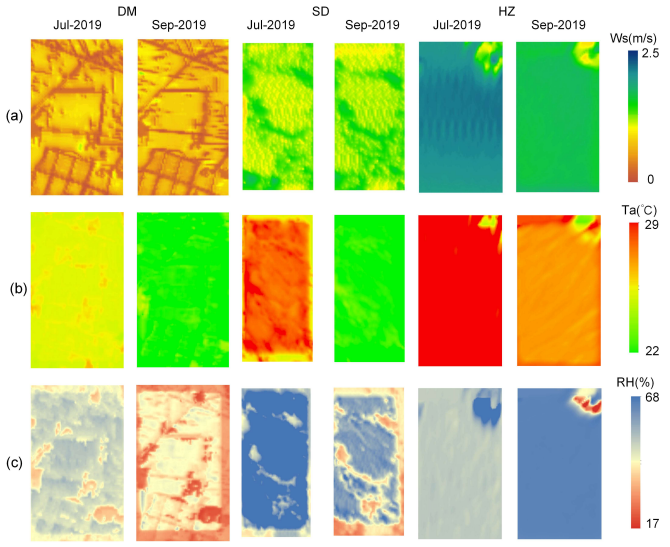

Fig. 6. Spatial distributions of CFD-based (a) wind speed, (b) Ta, and (c) RH, respectively.

showed the best agreement among the three variables, with an MRE of only 0.58% under both dry and wet conditions [see Fig. 5(c)].

The spatial distributions of wind speed, Ta, and RH from the CFD model for July and September 2019 are shown in Fig. 6. As expected, there are differences in meteorological data across individual pixels. These distributions exhibited spatial patterns similar to those of the underlying surfaces to some extent [see Fig. 2(c)–(e)], contrasting sharply with the homogeneous assumptions derived from site-based meteorological observations. The croplands, shelterbelts, residential areas, and reeds exhibited a noticeable wind shielding effect with reduced wind speeds. The vegetated surfaces also had stronger cold and wet island effects than the surrounding roads, residential areas,

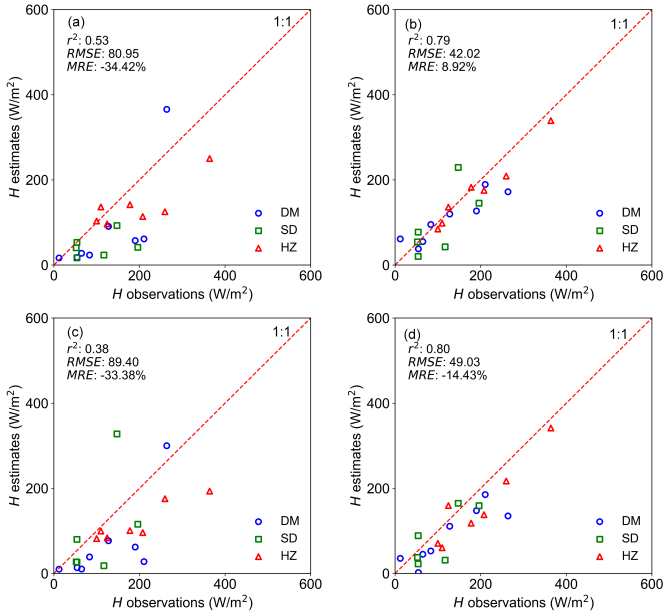

Fig. 7. Validation of H estimates derived from four different combination schemes of (a) PHY_Site, (b) ML_Site, (c) PHY_CFD, and (d) ML_CFD with EC-based H observations.

and deserts. Differences among the three experimental fields were well captured, with the HZ experimental field exhibiting higher wind speeds and Tas, and lower humidity than the other two experimental fields. Seasonal variations in Ta and RH also revealed that high temperatures and humidity levels typically occurred during the summer months (i.e., July).

B. Comparison of the Estimated and Observed Surface Fluxes

Once the downscaled meteorological data, z_{0m}, and kB^{-1} are determined, the spatially distributed H and LE can be estimated using the SEBS model. In addition to the scheme using CFD model-based meteorological data and ML-based z_{0m} and kB^{-1}, we also adopted site-based meteorological data and z_{0m} and kB^{-1} estimates from the physical parameterization schemes to drive the SEBS model. Four combination schemes with different meteorological data and estimation schemes were developed in this study. These four schemes are categorized as follows: schemes using site-based meteorological data and physical parameterization schemes (PHY_Site), site-based meteorological data and ML-based estimation schemes (ML_Site), CFD-based meteorological data and the physical parameterization scheme (PHY_CFD), and CFD-based meteorological data and ML-based estimation scheme (ML_CFD). The pixels located within the source area of the EC were weighted and averaged for comparison with the corresponding observations [11], [18].

Fig. 7 presents the validation results of H estimates derived from four different combination schemes. In general, most combination schemes showed varying degrees of underestimation, with MRE values less than −30% observed in the two physical parameterization schemes. The RMSE values achieved by the two ML-based schemes were 42.02 W/m^2

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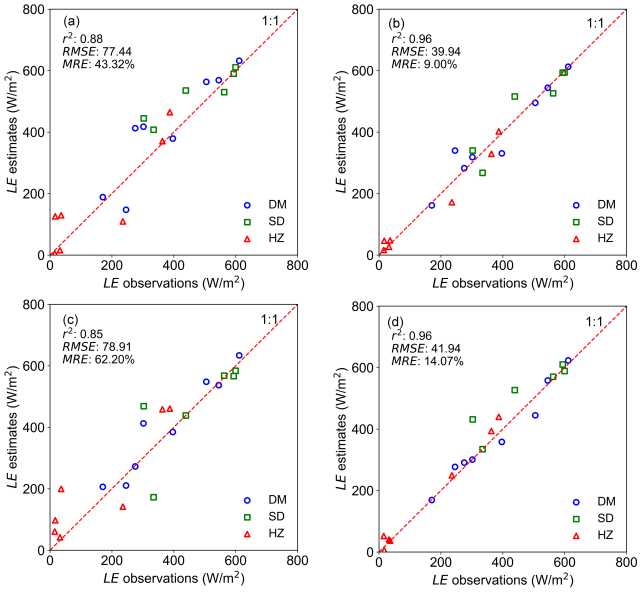

Fig. 8. Validation of LE estimates derived from four different combination schemes of (a) PHY_Site, (b) ML_Site, (c) PHY_CFD, and (d) ML_CFD with EC-based H observations.

for the site-based meteorological data and 49.03 W/m^2 for the CFD-based meteorological data, which were significantly lower than those of the two physically based schemes, with RMSE values of 80.95 and 89.40 W/m^2, respectively. Therefore, the performance of the CFD-based meteorological schemes is comparable to that of site-based data. The comparison of the four combination schemes also reveals similarities in performance within the two ML-based schemes and within the two physical-based schemes.

The validation results of the LE estimates derived from the four combination schemes are presented in Fig. 8. As expected, the performance ranking is consistent with that observed in the validation of H estimates. The estimates of LE for the four combination schemes exhibited a general trend of overestimation to different degrees. The two ML-based combination schemes outperformed the physical-based combination schemes in terms of the RMSE and MRE statistical metrics. The RMSE values for PHY_Site, PHY_CFD, ML_Site, and ML_CFD for the LE estimates were 77.44, 78.91, 39.34, and 43.32 W/m^2, respectively, whereas the MRE values were 43.32%, 62.20%, 9.00%, and 14.07%, respectively. The validation of LE estimates confirmed again that the two ML-based combination schemes performed better than the physical parameterization scheme. Compared to site-based meteorological observations, the use of CFD model-based meteorological data resulted in comparable performance.

In addition to validation against EC observations, we also compared our results with observations from an OMS installed at the DM experimental field in 2020. As shown in Fig. 2(c), the footprint of the OMS was significantly larger than that of the EC, encompassing a broader range of underlying surfaces, such as shelterbelts, roads, and residential areas. The pixels within the OMS footprint were weighted and averaged for comparison with OMS observations, and the

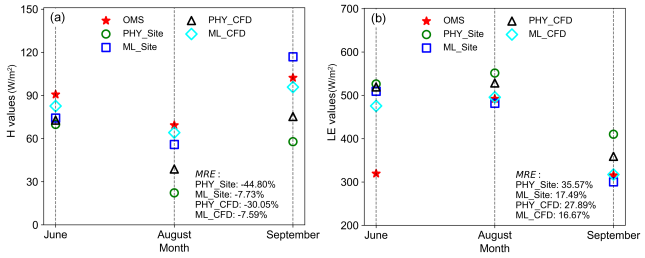

Fig. 9. Comparison of observed LE and H from OMS systems with four combination schemes for (a) H estimates and (b) LE estimates in the year 2020 from the DM experimental field.

validation results are presented in Fig. 9. Due to the limited number of observations, we present only the statistical metric of MRE for comparison purposes. Based on MRE, the two ML-based combination schemes showed better agreement with OMS observations for both H and LE estimates than the physical-based combination schemes. Specifically, the ML_Site and ML_CFD schemes yielded MRE values of −7.73% and −7.59% for H estimates, and 17.49% and 16.67% for LE estimates, respectively. In contrast, the Phy_Site and Phy_CFD schemes exhibited reduced accuracy, with MRE values of −44.80% and −30.05% for H estimates, and 35.57% and 27.89% for LE estimates, respectively. Notably, the combination schemes using CFD-based meteorological data demonstrated improved accuracy over those using site-based meteorological data. This finding is slightly different from the validation results against EC observations. On average, mean MRE values for CFD-based meteorological data were 18.82% for H estimates and 22.28% for LE estimates, while site-based meteorological data yielded MRE values of 26.27% for H estimates and 26.53% for LE estimates. These results highlight the importance of downscaled meteorological data when the representativeness of site-based meteorological observations is insufficient over large and heterogeneous underlying surfaces.

C. Spatial and Temporal Patterns of UAV-Based LE

Figs. 10 and 11 show the spatial distributions of LE derived from the four combination schemes and related auxiliary variables across an area of interest measuring 350 × 500 m within the three experimental fields. Regardless of the combination scheme used, the spatial distribution of LE varied significantly among the different land surface types within each experimental field. In the DM experimental field, areas characterized by built-up and barren land exhibited relatively low ET compared to those of croplands, orchards, and shelterbelts [see Fig. 10(a)]. In the SD experimental field, reed and water areas exhibited high LE values due to their well-watered conditions [see Fig. 10(b)], with relatively low land Ts, as shown in Fig. 11(b). Conversely, the roads in the SD experimental field had the lowest LE. In the HZ experimental field, the cropland situated in the northeast corner exhibited the highest LE compared to areas covered by *Kalidium foliatum* and deserts [see Fig. 10(c)]. The spatial pattern in the HZ experimental field also reveals microtopographic effects, with higher LE values observed in topographic valleys where

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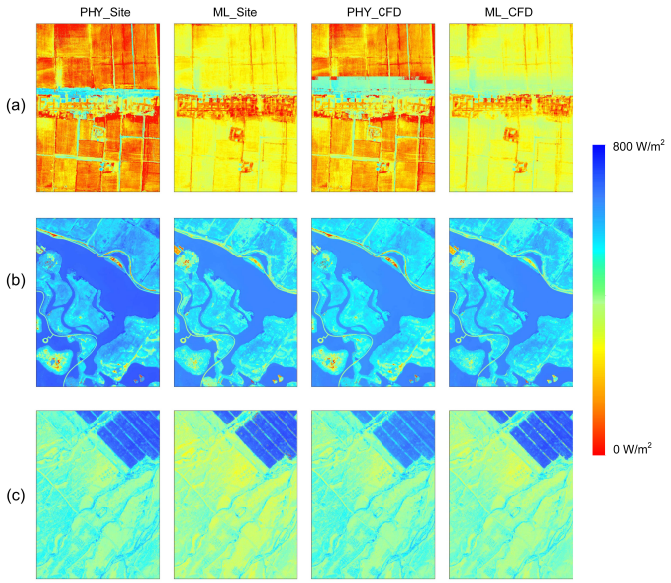

Fig. 10. Spatially distributed LE derived from four combination schemes over an area of 350 × 500 m for (a) DM, (b) SD, and (c) HZ experimental fields. The date for the DM experimental field is May 27, 2019, for the SD experimental field is June 30, 2019, and for the HZ experimental field is June 28, 2019, respectively.

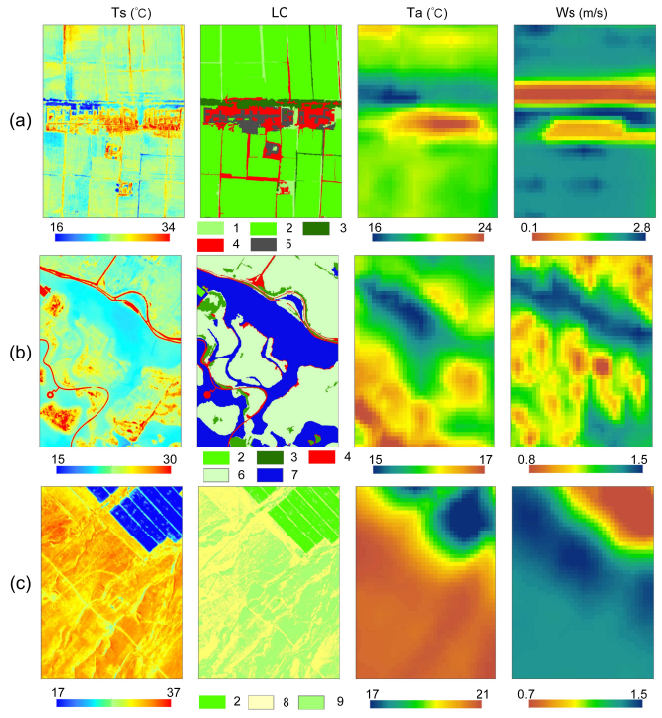

Fig. 11. Spatially distributed auxiliary variables (i.e., Ts, LC, Ta, and Ws) over an area of 350 × 500 m for (a) DM, (b) SD, and (c) HZ experimental fields. The numbers from 1 to 9 in LC represent grassland, maize, shelterbelt, road, built-up areas, reed, water, desert, and *Kalidium foliatum* in ascending order. The dates for the DM, SD, and HZ experimental fields are May 27, 2019, June 6, 2019, and June 6, 2019, respectively.

Kalidium foliatum and desert prevail, and lower values in topographic peaks.

In terms of different combination schemes, the spatial patterns achieved by PHY_Site are consistent with those

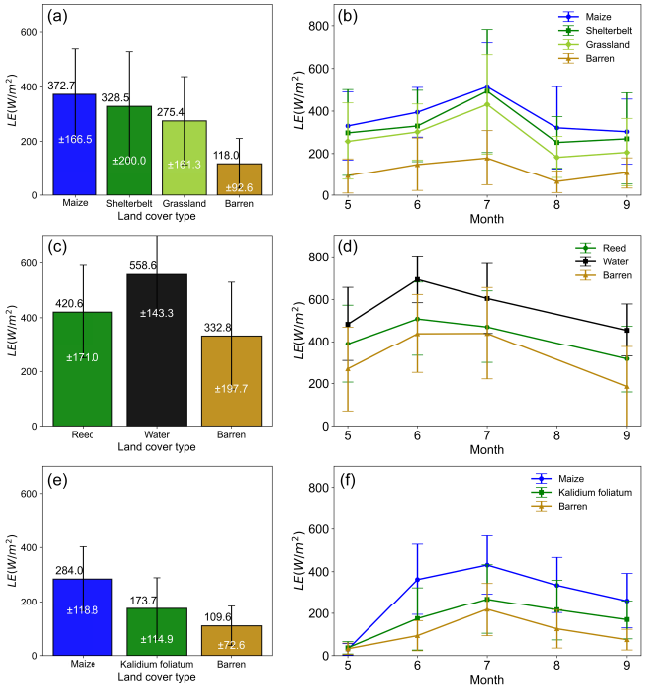

Fig. 12. Characteristics and temporal variations of LE under different LC types during the year 2019 for (a) and (b) DM experimental field, (c) and (d) SD experimental field, and (e) and (f) HZ experimental field.

achieved by PHY_CFD, whereas the patterns derived from ML_Site are similar to those from ML_CFD across the three experimental fields [see Fig. 10(a)–(c)]. The spatial patterns of the meteorological data highlight the significant impact of vegetated underlying surfaces on reducing Ta and wind speed [see Fig. 11(a)–(c)], contributing to spatial differences within the same estimation schemes when different meteorological data are used. The influence of meteorological data on LE spatial patterns was minimal in the HZ experimental field, with a relatively homogeneous spatial pattern, as shown in Fig. 11(c). However, in the DM experimental field, the differences in meteorological data are more obvious to the north of the shelterbelt, as Fig. 11(a) shows weakened southerly winds due to shelterbelt drag and the formation of a low-temperature band.

Regarding the performance of four combination schemes, we used the ML_CFD to present the temporal variations in LE across the three experimental fields (see Fig. 12). As expected, LE was highest in the SD experimental field because of its extensive water bodies, with mean values ranging from 332.8 W/m^2 (barren) to 558.6 W/m^2 (water) [see Fig. 12(c)]. In contrast, the HZ experimental field exhibited the lowest values due to its vast desert areas, with mean values ranging from 109.6 W/m^2 (desert) to 284.0 W/m^2 (maize) [see Fig. 12(e)]. The maize in the DM experimental field benefited from frequent irrigation, which maintained adequate soil moisture, resulting in relatively high mean values (327.7 W/m^2) compared to grassland and barren land [see Fig. 12(a)]. The same LC types also showed large standard deviations, implying the influence of water conditions [18]. Shelterbelt, barren, and maize exhibited the largest standard deviations, with values of ±200, ±197.7, and ±118.8 W/m^2 in the DM, SD, and

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HZ experimental fields, respectively [see Fig. 12(a), (c), and (e)]. In terms of temporal variation in LE, all LC types showed distinct seasonal characteristics, with peak LE values observed in June and July. The temporal variation in standard deviation also highlighted higher LE values predominantly in July, reflecting increased ET. The lowest standard deviations were recorded in August, September, and May for the DM, SD, and HZ experimental fields [see Fig. 12(b), (d), and (f)], indicating high consistency with the months when LE reached its lowest value.

V. Discussion

Given the background of climate change and biodiversity loss, accurate fine-scale monitoring of ET is crucial for managing risks to land and water ecosystems [3], [4]. UAVs offer a significant opportunity to monitor water and heat fluxes at fine scales. However, a key challenge lies in the spatial resolution of UAV imagery, which is typically less than 1 m. This necessitates comprehensive consideration in ET estimation models and corresponding parameterization schemes [14]. In addition, it is often assumed that the underlying surface within the flying area is homogeneous, and meteorological data from nearby weather stations are used to drive ET estimation models. This assumption hinders operational applications in regions lacking meteorological observations or in those with high land surface heterogeneity. Addressing these challenges is essential for achieving accurate estimates of water and heat fluxes based on UAV data, particularly under heterogeneous land surface conditions during flying campaigns.

In this study, we employed ML methods to bypass the complexities involved in parameterizing z_{0m} and kB^{-1}. Typically, the physically based parameterization schemes for z_{0m} rely on critical but unmeasured parameters, such as the drag coefficient for roughness elements and the vertical foliage distribution [20], [30]. This reliance on uncertain or unavailable parameters can result in substantial estimation errors, as evidenced by the 377.65% overestimation observed for the physical parameterization scheme in Fig. 3(d). The use of ML for alternative estimation schemes significantly improved the accuracy of z_{0m} and kB^{-1} estimates compared to the physical parameterization scheme, as presented in Fig. 3(a) and (b). Hu et al. [30] previously evaluated ML using the FLUXNET2015 dataset, achieving RMSE values of 0.23 m for z_{0m} and 1.97 for kB^{-1} at the site scale. In contrast, the RMSE values obtained in this study were 0.04 m for z_{0m} and 2.41 for kB^{-1}, which are comparable to those reported in the previous study.

In terms of estimating water and heat fluxes, the ML-based z_{0m} and kB^{-1} have demonstrated robustness in estimating H and LE using both site- and CFD-based meteorological data compared with the physical-based z_{0m} and kB^{-1}. The SEBS model has previously tended to overestimate LE and underestimate H [26], [55], as clearly shown in Figs. 7 and 8. These deviations were mitigated by optimizing z_{0m} and kB^{-1} using ML methods. Specifically, the RMSE values for H estimates decreased from 80.92 to 42.02 W/m^2 and from 89.40 to 49.03 W/m^2 for LE estimates when site-based meteorological data with the ML-based estimation schemes were used. Similarly, with the CFD-based meteorological input data, the RMSE values improved from 78.91 to 41.94 W/m^2 for the LE estimates and from 77.44 to 39.94 W/m^2 for the H estimates. Furthermore, validation against OMS with a larger footprint with diverse LC types also revealed similar accuracy improvements (see Fig. 9). These findings demonstrate that the ML-based alternative estimation schemes have significantly improved the overall performance in estimating water and heat fluxes.

In addition to adopting the ML method, this study also utilized meteorological data with a spatial resolution of 30 m from the CFD model. Although the flying areas of the three experimental fields covered only 2 $\times$ 1 km, the relative heterogeneity of the land surface resulted in variations in meteorological variables across individual pixels. The vegetated underlying surfaces exhibited significant wind shielding, wetting, and cooling effects, as shown in Fig. 6. The spatial distributions of the meteorological data from the CFD models revealed notable differences compared to those obtained using only site-based observations, as detailed in Table I. This variation arises because the underlying surfaces in the study area are diverse, covering a larger area than those typically reported in previous studies [4], [13]. The complexity of local meteorological conditions is largely driven by the heterogeneity of underlying surfaces [38]. This diversity explains why the HZ experimental field exhibited relatively uniform meteorological conditions, dominated by underlying surface types such as deserts and *Kalidium foliatum*, with low roughness element height and low ET.

Validation against EC observations at the footprint scale showed that the downscaled meteorological data for estimating H and LE yielded accuracy comparable to that achieved with site-based observations. There is a slight decrease in RMSE values (less than 9 W/m^2) for both physical- and ML-based H and LE estimates when downscaled meteorological data are used. This minor difference can be attributed to the consistency of the downscaled meteorological data with the EC observation footprint, as depicted in Fig. 5. Validations against H and LE observations from the OMS revealed a slight improvement in accuracy, as shown in Fig. 9. Both physical- and ML-based SEBS models that use CFD-based meteorological data achieved lower MREs (18.82% for H estimates and 22.28% for LE estimates) compared to those using site-based meteorological data (26.27% for H estimates and 26.53% for LE estimates). This improvement stems from the larger source area covered by the OMS compared to the EC in the DM experimental area, resulting in more complex meteorological conditions [52]. Therefore, the CFD-based meteorological data can also be considered for applications where the flying area is large and meteorological conditions cannot be adequately represented by a single site. More importantly, the downscaled meteorological data from CFD models can be operationally applied in regions lacking site-based meteorological observations [14].

In general, the use of ML-based alternative estimation schemes significantly improved the performance of H and LE estimates in this study. It is important to assess the accuracy achieved in previous studies of UAV-based ET estimation. In

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TABLE III
SUMMARY OF PREVIOUS STUDIES USING UAV TO ESTIMATE LE

Model used	Main vegetation type	Observed mean LE (W/m^2)	Validation instruments	RMSE for LE (W/m^2)	Relative RMSE (%)	References
TSEB/ DTD	Barley, gravel road, conifers	255	EC system	94 / 67	37%/ 26%	Hoffmann et al. [16]
TSEB	Vineyard, grass	217-293	EC system	58-77	20%-35%	Xia et al. [5]
3T	Arbors, lawns, urban infrastructure, water bodies	154	EBBR system	30	20%	Qin et al. [17]
TSEB/DTD	Grass, tree	380-425	EC system	60	14%-16%	Simpson et al. [3]
OSEB	Grass	200-220	EC system	45	21%-23%	Morgan et al. [14]
SEBS	Maize, reed, desert	320	EC system	70	22%	Wei et al. [18]
SEBS (ML_CFD)	Maize, reed, desert	320	EC system	42	13%	**This study**

Note: OSEB is the optimal surface balance model, 3T is the three-temperature model, EBBR is the energy balance Bowen ratio.

our previous study, the optimization of kB^{-1} using a combined physical parameterization scheme resulted in RMSE values of 53 W/m^2 for H and 70 W/m^2 for LE estimates [18]. This study using the ML-based alternative estimation schemes achieved slightly lower RMSE values of 42.02 and 49.03 W/m^2 for H with different meteorological data inputs and of 39.94 and 41.94 W/m^2 for LE with different meteorological data inputs. To further demonstrate the accuracy of LE estimation in this study, we summarized the validation results reported in recent UAV-based LE estimation studies, as shown in Table III. Given the variation in vegetation types and meteorological conditions across these studies, we calculated the relative RMSE based on the reported RMSE values and the corresponding observed mean LE values to enable a more consistent comparison. The relative RMSEs reported in previous studies ranged from 14% to 35% [3], [14], [16], [17], [18]. In comparison, our proposed model demonstrates generally satisfactory performance, with the ML_CFD scheme achieving the lowest relative RMSE of 13%. This further underscores the robustness and competitiveness of the model developed in this study.

Although the ML-based alternative estimation schemes proposed in this study achieved high accuracy, some uncertainties remain, particularly the risk of overfitting, where model performance degrades from training to validation data [57]. To mitigate this, we incorporated as many sites as possible across the three UAV campaigns and applied strict data quality controls [30], [31]. Hyperparameters were optimized via grid search, and model performance was evaluated using tenfold cross-validation to enhance robustness and reduce overfitting risks. When training data are limited, incorporating physical constraints such as domain knowledge, conservation laws, or hybrid physical-ML structures can improve the physical consistency of ML outputs and reduce overfitting. This approach has been widely adopted in temperature and soil moisture simulations [58], [59], [60]. In regions lacking site-based observations, training ML models becomes more challenging. Recently, transfer learning has shown strong potential in fields such as LC classification and vegetation monitoring [61]. Following established strategies, models can be pretrained using large-scale datasets (e.g., FLUXNET) and fine-tuned with site-specific data from areas sharing similar meteorological and surface conditions [56], [61]. This enhances model adaptability and improves estimation accuracy in data-scarce environments. Although transfer learning may slightly reduce performance on the source domain, it offers a practical and scalable solution for extending ML applications to regions with limited or no in situ data.

VI. CONCLUSION

In this study, we propose a robust framework for improving fine-scale ET estimation using UAV-based multispectral and thermal images. Our proposed ML-based alternative estimation schemes achieved nearly a twofold improvement in the accuracy of key parameter estimates compared to physical parameterization schemes. The RMSE of z_{0m} estimates improved from 0.07 to 0.04 m, and for kB^{-1} estimates, it improved from 4.58 to 2.41. The CFD-based downscaled meteorological data also showed good agreement with site-based observations, with RMSE values for Ws, Ta, and RH estimates of 0.63 m/s, 1.47 °C, and 5.87%, respectively. These two improvements enable us to obtain relatively accurate key parameter estimates and reduce reliance on site-based meteorological data.

The ML-based alternative estimation schemes demonstrated improved accuracy, with H estimates improving from 80.95 to 42.02 W/m^2 and LE estimates improving from 77.44 to 39.94 W/m^2, compared to physical parameterization schemes using site-based meteorological data. The use of CFD-based downscaled meteorological data also resulted in accuracy comparable to that of site-based meteorological data. In particular, CFD-based downscaled meteorological data are superior to site-based meteorological data for H and LE estimates over large and heterogeneous underlying surfaces. Compared with site-based meteorological data as inputs, the mean MRE values were improved from 26.27% to 18.82% for H estimates and 26.53%–22.28% for LE estimates when using CFD-based meteorological data. While we use site-based flux observations within the flying area to train ML-based alternative estimation schemes, integrating publicly available FLUXNET site-based observations with similar geographic and meteorological conditions for transfer learning could help reduce this reliance [56].

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ACKNOWLEDGMENT

The authors would like to thank all the scientists, engineers, and students who participated in the HiWATER-MUSOEXE, MUSOES-UAV, and integrated observation network of surface processes in the Heihe River Basin (HRB). They would like to thank the Center for Geodata and Analysis, Faculty of Geographical Science, Beijing Normal University, for the high-performance computing support. They gratefully acknowledge Xuelong Chen from the Institute of Tibetan Plateau Research, Chinese Academy of Sciences, for providing the surface energy balance system (SEBS) code (https://github.com/TSEBS/SEBS_Spain). The extreme gradient boosting (XGBoost) code is publicly available at https://github.com/dmlc/xgboost. The MUSOES-UAV dataset is currently undergoing peer review for public dissemination. To maintain the integrity of the evaluation process, the data will be published through the National Tibetan Plateau Data Center (http://data.tpdc.ac.cn) upon completion of the peer-review process.

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